ABO Blood Type Analysis

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- Introduction
- Data Analysis
- Conclusion

Introduction

Studying the correlation between ABO blood types and certain types of cancer is of particular interest due to the potential implications of such genetic factors' influence on cancer susceptibility. The ABO blood group system, which is determined by antigens on the surface of red blood cells has been extensively studied for its essential role in blood transfusions and demonstrates a potential to be correlated with other health conditions, including cancer. Recent epidemiologic studies have established curious patterns of certain blood types having a higher or lower risk of developing particular types of cancer. Some studies have suggested that individuals with Type A blood have a higher risk of, for instance, stomach cancer when compared to other blood types. At the same time, studies on Type O blood have indicated that some cancers have a lower risk among people with this blood type. This project plans to study these correlations closely by using data on blood types and cancer incidence rates to determine if the presence of certain antigens that are directly related to the ABO blood type influence cancer development . This study's results may potentially present an important contribution to understanding cancer etiology and may also propose directions for future research on cancer prevention and treatment based on genetic predispositions.

The data used in this project was gathered from this link: https://ispub.com/IJPA/13/1/5982 (https://ispub.com/IJPA/13/1/5982)

Data Analysis

In this project, we will conduct a χ^2 test of independence to analyze the association between blood type in cancer and control group. We would use $\alpha = 0.05$ as a critical value. Should significant differences emerge, we will subsequently examine each specific cancer type in comparison to the control group. This analysis aims to identify which cancer type contributes most significantly to these differences, considering both blood genotype and phenotype.

We will first proceed onto the phenotype analyzation.

Our hypothesis for the entire project is consistent, that our H_0 is There is no association between cancer presence (cancer vs. no cancer) and blood genotype (or phenotype) in the population. Our H_A is There is an association between cancer presence (cancer vs. no cancer) and blood genotype (or phenotype) in the population.

```
control_counts <- c(A = 490, B = 1070, AB = 300, O = 780)
total_counts <- c(A = 903, B = 1068, AB = 246, O = 423)
contingency_table <- matrix(c(control_counts, total_counts), nrow = 2, byrow = TRUE)
contingency_table
```

```
##      [,1] [,2] [,3] [,4]
## [1,]  490 1070  300  780
## [2,]  903 1068  246  423
```

```
chisq.test(contingency_table)
```

```
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 233.73, df = 3, p-value < 2.2e-16
```

Our p-value is extremely close to 0. So we reject H_0 and conclude that there is a correlation between blood phenotypes and cancers. Therefore, we should move forward with comparing individual cancer types against the control group. It should also be noted that there have been typographical discrepancies within the data set. We are operating under the assumption that the total counts reflect the accurate data, and the contingency tables have been adjusted accordingly to align with the corrected totals. Please refer to the comments in the code for the adjustment details.

```
cancer_counts <- matrix(c(
  196, 210, 42, 112,
  49, 84, 21, 56,
  14, 35, 4, 17,
  84, 56, 21, 7,
  196, 140, 28, 98,
  112, 154, 84, 56,
  98, 154, 21, 21,
  28, 42, 2, 12,  #it was 4 in the original data, but should be 82
  42, 105, 14, 28,
  21, 5, 2, 7,    #it was 12, corrected to be 5
  28, 20, 0, 2,
  35, 63, 7, 7
), nrow = 12, byrow = TRUE)
cancer_types <- c("Oral cavity", "Gastrointestinal tract", "Gall bladder", "Lungs", "Breast",
"Cervix", "Ovary", "Lymphoma", "Bone", "Kidney", "Urinary bladder", "Central nervous system")
chi_square_results <- list()
for (i in 1:nrow(cancer_counts)) {
  type_counts <- cancer_counts[i, ]
  contingency_table <- matrix(c(control_counts, type_counts),nrow = 2, byrow = TRUE)
  chi_square_test <- chisq.test(contingency_table)
  chi_square_results[[cancer_types[i]]] <- chi_square_test
}
```

```
## Warning in chisq.test(contingency_table): Chi-squared approximation may be
## incorrect
```

```
chi_square_results
```

```
## $`Oral cavity`
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 80.857, df = 3, p-value < 2.2e-16
##
##
## $`Gastrointestinal tract`
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 3.2272, df = 3, p-value = 0.3579
##
##
## $`Gall bladder`
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 4.1573, df = 3, p-value = 0.245
##
##
## $Lungs
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 114.89, df = 3, p-value < 2.2e-16
##
##
## $Breast
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 131.9, df = 3, p-value < 2.2e-16
##
##
## $Cervix
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 71.186, df = 3, p-value = 2.378e-15
##
##
## $Ovary
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 90.651, df = 3, p-value < 2.2e-16
##
```

```
##
## $Lymphoma
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 23.576, df = 3, p-value = 3.062e-05
##
##
## $Bone
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 26.731, df = 3, p-value = 6.703e-06
##
##
## $Kidney
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 39.018, df = 3, p-value = 1.721e-08
##
##
## $`Urinary bladder`
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 52.42, df = 3, p-value = 2.438e-11
##
##
## $`Central nervous system`
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 38.425, df = 3, p-value = 2.298e-08
```

Except for the gastrointestinal tract and gall bladder, all other cancer types show associations with blood phenotype, with p-values having at least five zeros after the decimal point, suggesting significant differences. Now, we will proceed onto the analysis on blood genotype.

```
ldhw<-function(nA=1, nB=2, nAB=3, nO=4, tol=0.01) {
  pA <- seq(0,1, by=tol)
  pB <- seq(0,1, by=tol)
  lz <- matrix(0, nrow=length(pA), ncol=length(pB))
  n <- nA+nB+nO+nAB
  for(i in 1:length(pA)) {
    for(j in 1:length(pB) ) {
      a <- pA[i]
      b <- pB[j]
      if( a+b < 1) {
        c <- 1-a-b
        lsA <- log(a^2 + 2*a*c)
        lsB <- log(b^2 + 2*b*c)
        lsO <- log(c^2)
        lsAB <- log(2*a*b)
        lz[i,j] <- lsA*nA + lsB*nB + lsO*nO + lsAB*nAB
      } else {
        lz[i,j] <- -Inf
      }
    }
  }
  ijmax <- which.max(lz)
  imax <- 1+ ijmax%%length(pA)
  jmax <- 1+ ijmax%/%length(pA)
  pAmax <- pA[imax]
  pBmax <- pB[jmax]
  pOmax <- 1-pAmax-pBmax
  #print("Approximate MLE: (pA,pB,pO) = ")
  #print( c(pAmax,pBmax,pOmax))
  matrix_result<-2640*c(pAmax,pBmax,pOmax)*2
  #print("Approximate Counts = ")
  #print(matrix_result)
  return(matrix_result)
}
```

```
control_counts<-ldhw(nA=490, nB=1070, nAB=300, nO=780)
total_counts<-ldhw(nA=903, nB=1068, nAB=246, nO=423)
contingency_table <- matrix(c(control_counts, total_counts),
                            nrow = 2, byrow = TRUE)
chisq.test(contingency_table)
```

```
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 165.88, df = 2, p-value < 2.2e-16
```

Again, the test showed statistically significant evidence that blood genotype is associated with cancers. We will check each individual cancer against the control group.

```
cancer_counts<-ldhw(nA=196, nB=210, nAB=42, nO=112)
contingency_table <- matrix(c(cancer_counts, total_counts),
                            nrow = 2, byrow = TRUE)
print("Oral cavity:")
```

```
## [1] "Oral cavity:"
```

```
chisq.test(contingency_table)
```

```
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 30.587, df = 2, p-value = 2.28e-07
```

```
cancer_counts<-ldhw(nA=49, nB=84, nAB=21, nO=56)
contingency_table <- matrix(c(cancer_counts, total_counts),
                            nrow = 2, byrow = TRUE)
print("Gastrointestinal tract:")
```

```
## [1] "Gastrointestinal tract:"
```

```
chisq.test(contingency_table)
```

```
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 109.41, df = 2, p-value < 2.2e-16
```

```
cancer_counts<-ldhw(nA=14, nB=35, nAB=4, nO=17)
contingency_table <- matrix(c(cancer_counts, total_counts),
                            nrow = 2, byrow = TRUE)
print("Gall bladder:")
```

```
## [1] "Gall bladder:"
```

```
chisq.test(contingency_table)
```

```
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 230.18, df = 2, p-value < 2.2e-16
```

```
cancer_counts<-ldhw(nA=84, nB=56, nAB=21, nO=7)
contingency_table <- matrix(c(cancer_counts, total_counts),
                            nrow = 2, byrow = TRUE)
print("Lungs:")
```

```
## [1] "Lungs:"
```

```
chisq.test(contingency_table)
```

```
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 298.05, df = 2, p-value < 2.2e-16
```

```
cancer_counts<-ldhw(nA=196, nB=140, nAB=28, nO=98)
contingency_table <- matrix(c(cancer_counts, total_counts),
                            nrow = 2, byrow = TRUE)
print("Breast:")
```

```
## [1] "Breast:"
```

```
chisq.test(contingency_table)
```

```
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 112.86, df = 2, p-value < 2.2e-16
```

```
cancer_counts<-ldhw(nA=112, nB=154, nAB=84, nO=56)
contingency_table <- matrix(c(cancer_counts, total_counts),
                            nrow = 2, byrow = TRUE)
print("Cervix:")
```

```
## [1] "Cervix:"
```

```
chisq.test(contingency_table)
```

```
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 56.828, df = 2, p-value = 4.569e-13
```

```
cancer_counts<-ldhw(nA=98, nB=154, nAB=21, nO=21)
contingency_table <- matrix(c(cancer_counts, total_counts),
                            nrow = 2, byrow = TRUE)
print("Ovary:")
```

```
## [1] "Ovary:"
```

```
chisq.test(contingency_table)
```

```
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 98.794, df = 2, p-value < 2.2e-16
```

```
cancer_counts<-ldhw(nA=28, nB=42, nAB=2, nO=12)
contingency_table <- matrix(c(cancer_counts, total_counts),
                            nrow = 2, byrow = TRUE)
print("Lymphoma:")
```

```
## [1] "Lymphoma:"
```

```
chisq.test(contingency_table)
```

```
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 36.882, df = 2, p-value = 9.801e-09
```

```
cancer_counts<-ldhw(nA=42, nB=105, nAB=14, nO=28)
contingency_table <- matrix(c(cancer_counts, total_counts),
                            nrow = 2, byrow = TRUE)
print("Bone:")
```

```
## [1] "Bone:"
```

```
chisq.test(contingency_table)
```

```
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 171.09, df = 2, p-value < 2.2e-16
```

```
cancer_counts<-ldhw(nA=21, nB=5, nAB=2, nO=7)
contingency_table <- matrix(c(cancer_counts, total_counts),
                            nrow = 2, byrow = TRUE)
print("Kidney:")
```

```
## [1] "Kidney:"
```

```
chisq.test(contingency_table)
```

```
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 663.33, df = 2, p-value < 2.2e-16
```

```
cancer_counts<-ldhw(nA=28, nB=20, nAB=0, nO=2)
contingency_table <- matrix(c(cancer_counts, total_counts),
                            nrow = 2, byrow = TRUE)
print("Urinary bladder:")
```

```
## [1] "Urinary bladder:"
```

```
chisq.test(contingency_table)
```

```
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 146.05, df = 2, p-value < 2.2e-16
```

```
cancer_counts<-ldhw(nA=35, nB=63, nAB=7, nO=7)
contingency_table <- matrix(c(cancer_counts, total_counts),
                            nrow = 2, byrow = TRUE)
print("Central nervous system:")
```

```
## [1] "Central nervous system:"
```

```
chisq.test(contingency_table)
```

```
##
##  Pearson's Chi-squared test
##
## data:  contingency_table
## X-squared = 165.82, df = 2, p-value < 2.2e-16
```

For genotype, every test showed a significant difference, meaning that genotype is associated with all types of cancers.

Conclusion

To summarize our findings, both blood genotype and phenotype demonstrated associations, indicating that certain blood alleles may be more susceptible to specific cancers, with the exception of the gastrointestinal tract and gall bladder for phenotype. However, our primary goal in this project is to utilize the chi-square test for independence to assess the relationship between blood type and cancer (either individually or collectively), so the test does not allow us to determine which specific blood types are more vulnerable to particular cancers. For more detailed insights, further analyses such as ANOVA followed by post hoc tests could be conducted using the same data set.